MACHINE LEARNING · MEDICAL IMAGING

Adriano Pinto

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PROFESSIONAL SUMMARY

I am a researcher and data scientist drawn to multidisciplinary challenges, a path that took me from Biomedical Engineering to a PhD in Machine Learning, specializing in Deep Learning for medical imaging such as stroke, cardiac MR, and brain tumour segmentation. At heart I remain an individual contributor: I like to read and follow the literature, prototype ideas, and turn technology into real clinical applications.

PROFESSIONAL EXPERIENCE

Senior AI Researcher

iLoF

Jan 2026 — Ongoing · Portugal · Hybrid

- Contribute to research and development of ML-based frameworks for classification of RAMAN and FTIR spectra.
- Mentoring a brief team of data scientists, fostering innovation and collaboration.

Head of AI

AI4MedImaging

Oct 2024 — Dec 2025 · Portugal · Hybrid

- Lead research and product teams, defining critical features and driving innovation.
- Propose and oversee new research initiatives and product enhancements.
- Remain hands-on, contributing to the development and implementation of AI models.

Senior Data Scientist

AI4MedImaging

Feb 2022 — Oct 2024 · Portugal · Hybrid

- Researched and developed AI-driven solutions focused on deep learning for segmentation and classification tasks, aligned with user requirements.

Data Scientist

DTx CoLab

Mar 2020 — Jan 2022 · Guimarães, Portugal

- End-to-end machine learning solutions, pipelines, and software packages, focused on anomaly detection, computer vision, and time-series analysis for industry stakeholders.

Invited Assistant Professor

University of Minho

Sep 2018 — Ongoing · Braga, Portugal

- SEIMR (TP)
- Signal Processing (TP)
- Level 1 Electronics (Lab)

EDUCATION

Doctor of Philosophy

Informatics MAP-i

2015 — 2020

Research Internship

ISTB — University of Bern

Oct 2016 — Mar 2017 · Bern, Switzerland

Biomedical Engineering

Medical Electronics Branch

2010 — 2015 · University of Minho, Braga

PUBLICATIONS

Journal **Journal Articles**

1. S. Fernandes, C. Barros, **A. Pinto**, V. H. Pereira, C. Lima, C. A. Silva. "Modelling Stochastic Sensor Noise via Mask-Conditioned Diffusion for Data Augmentation in Low-SNR LGE-CMR." *Sensors*, vol. 26, no. 10, p. 2933, **2026**.
2. **A. Pinto**, J. Amorim, A. Hakim, V. Alves, M. Reyes, C. A. Silva. "Prediction of stroke lesion at 90-day follow-up by fusing raw DSC-MRI with parametric maps using deep learning." *IEEE Access*, vol. 9, pp. 26260–26270, **2021**.
3. **A. Pinto**, S. Pereira, R. Meier, et al. "Combining unsupervised and supervised learning for predicting the final stroke lesion." *Medical Image Analysis*, vol. 69, p. 101888, **2021**.
4. S. Winzeck, A. Hakim, R. McKinley, **A. Pinto**, V. Alves, C. A. Silva, M. Pisov, et al. "ISLES 2016 and 2017—Benchmarking Ischemic Stroke Lesion Outcome Prediction Based on Multispectral MRI." *Frontiers in Neurology*, vol. 9, **2018**.
5. S. Pereira, **A. Pinto**, J. Amorim, A. Ribeiro, V. Alves, C. A. Silva. "Adaptive feature recombination and recalibration for semantic segmentation with fully convolutional networks." *IEEE Transactions on Medical Imaging*, **2019**.
6. **A. Pinto**, R. McKinley, V. Alves, R. Wiest, C. A. Silva, M. Reyes, et al. "Stroke lesion outcome prediction based on MRI imaging combined with clinical information." *Frontiers in Neurology*, vol. 9, p. 1060, **2018**.
7. **A. Pinto**, S. Pereira, D. Rasteiro, C. A. Silva. "Hierarchical brain tumour segmentation using extremely randomized trees." *Pattern Recognition*, vol. 82, pp. 105–117, **2018**.
8. S. Pereira, **A. Pinto**, V. Alves, C. A. Silva. "Brain tumor segmentation using convolutional neural networks in MRI images." *IEEE Transactions on Medical Imaging*, vol. 35, no. 5, pp. 1240–1251, **2016**.
9. S. Pereira, **A. Pinto**, J. Oliveira, A. M. Mendrik, J. H. Correia, C. A. Silva. "Automatic brain tissue segmentation in MR images using Random Forests and Conditional Random Fields." *Journal of Neuroscience Methods*, vol. 270, pp. 111–123, **2016**.

Conference Proceedings

1. I. R. Lima, N. Leite, **A. Pinto**, P. M. Pires, C. Martins, N. Lopes. "eSardine: A General Purpose Platform with Autonomous AI and Explainable Outputs." *ICAART*, vol. 2, pp. 612–625, **2022**.
2. C. A. Silva, **A. Pinto**, S. Pereira, A. Lopes. "Multi-stage deep layer aggregation for brain tumor segmentation." *MICCAI BrainLes Workshop*, pp. 179–188, **2020**.
3. J. Amorim, **A. Pinto**, S. Pereira, C. A. Silva. "Segmentation squeeze-and-excitation blocks in stroke lesion outcome prediction." *IEEE ENBENG*, pp. 1–4, **2019**.
4. **A. Pinto**, S. Pereira, R. Meier, et al. "Enhancing clinical MRI perfusion maps with data-driven maps of complementary nature for lesion outcome prediction." *MICCAI 2018*, pp. 107–115, **2018**.
5. S. Pereira, **A. Pinto**, V. Alves, C. A. Silva. "Deep convolutional neural networks for the segmentation of gliomas in multi-sequence MRI." *BrainLes 2015*, Springer, pp. 131–143, **2015**.
6. **A. Pinto**, S. Pereira, H. Correia, J. Oliveira, D. M. Rasteiro, C. A. Silva. "Brain tumour segmentation based on extremely randomized forest with high-level features." *IEEE EMBC*, pp. 3037–3040, **2015**.
7. **A. Pinto**, S. Pereira, H. Dinis, C. A. Silva, D. M. L. D. Rasteiro. "Random decision forests for automatic brain tumor segmentation on multi-modal MRI images." *IEEE ENBENG*, pp. 1–5, **2015**.

Preprints

1. E. Costa, C. Barros, **A. Pinto**, N. Martins, L. Tondi, S. Pica, V. H. Pereira, M. Lombardi, et al. "Clinical validation of a fully automated framework for ventricular segmentation and wall motion classification from cine CMR." *Research Square*, preprint, **2022**.

TECHNICAL SKILLS & RESEARCH FOCUS

PROGRAMMING & TOOLING

Python · PyTorch · TensorFlow · Keras · Scikit-learn · Bash · Docker · Git · TeX

STRENGTHS

Continuous Learning · Critical Thinking · Problem Solving · Collaboration · Curiosity · Attention to Detail · Adaptability

AREAS OF INTEREST

Artificial Intelligence · Deep Learning · Medical Imaging · Healthcare · Research · Segmentation · Classification

Focus: *Deep Learning applied to medical imaging — stroke outcome prediction, cardiac MR, and brain tumour segmentation — from research through to deployable clinical AI products.*